# Experiment 1: pre-approved increase offer

Follows the ship decision in DECISIONS.pdf. Population: members with headroom (max eligible limit minus current limit of at least $100), about 77% of members.

> **Hypothesis:** once the slider is capped at a member's room, every request is approved, so asking members to *request* an increase adds friction and the feeling of a possible "no" for nothing. Presenting the same increase as a **pre-approved offer to accept** will raise adoption and add retention on top of the ~+6 pp the shipped flow already delivers.

## Background

In the first experiment, members with headroom retained 5.9 to 6.3 pp better at 30 days and 7.3 to 7.6 pp better at 60 days than control. Members already at their max lost 18 to 26 pp, because any request they made was declined. The shipped version shows the Increase button only to members with room and caps the slider at that room, so a decline is no longer possible. That shipped flow is arm A here.

## Arms

| | A: Request (shipped flow) | B: Pre-approved offer |
|---|---|---|
| Home | Increase button | Highlighted PRE-APPROVED card: "Raise your limit to $X. That's $Y more to work with. No application needed." Button: See your offer |
| Next screen | Request an increase: slider starts at $500 (or the cap if lower), capped at the member's room; Submit | You're pre-approved: new limit shown up front at the full amount, one-tap Accept. "Choose a smaller amount" reveals the same capped slider |
| Disclosure | None | Accepting runs a soft inquiry that won't affect the member's credit score; the new limit is ready right away |
| Decision | Always approved (amount is within room) | Always approved (amount is within room) |
| Result screen | Shared | Shared |

Members at their max and members who have already requested see no increase entry point in either arm.

## Design and metrics

| | |
|---|---|
| Assignment | Randomized 50/50 at signup among members with at least $100 of headroom; headroom is known at signup and fixed, so eligibility can't be influenced by the arm |
| Primary metric | 30-day retention (account open 30 days after joining), treatment B minus A, with 95% CI. 60-day retention read as follow-up |
| Secondary | Share of members who increase; days from joining to increase; increase taken as a share of available room |
| Guardrails | Delinquency and utilization on increased limits; support contacts about limits |
| Sample size | Arm A's 30-day retention is about 93.8%. Detecting a 1 pp difference at 80% power and 5% significance needs about 9,100 members per arm; the first experiment enrolled ~500k per arm, so enrollment of a few days is enough. The binding constraint is waiting 30 (and 60) days to read |
| Ship rule | Ship B if 30-day retention is higher with a CI excluding zero and guardrails are no worse than A; otherwise keep A |

## Try it in the app

The React Native app includes both arms. Start the mock API, then the web app, in two terminals:

```
cd mock-api && npm start
cd react_native_app && npm install && npm run web
```

- On Home, use **ARM** to switch between **A: Request** and **B: Pre-approved**, and **PERSONA** to switch members. The arm control is for demos; in production the arm comes from assignment.
- A member can request once, and the mock API keeps that in memory. Restart the mock API to reset all personas.

| Persona | Limit / max eligible | Room | What you should see (both arms) |
|---|---|---|---|
| approved | $2,500 / $5,000 | $2,500 | Entry point shown; accepting the full amount approves a $5,000 limit |
| utilization-too-high | $1,000 / $1,800 | $800 | Slider capped at $800; full amount approves a $1,800 limit |
| new-credit-lines | $500 / $1,000 | $500 | Slider capped at $500; full amount approves a $1,000 limit |
| insufficient-history | $1,500 / $1,500 | $0 | No Increase button or offer card |

## Implementation notes

- All changes are in `react_native_app/App.tsx`. No API changes: the app derives room as `max_eligible_limit_cents - limit_cents` from GET /v1/credit-line.
- The cap is the room rounded down to a $50 step, no higher than the API's $2,500 limit. The entry point is hidden when the cap is under the API's $100 minimum.
- Both arms share the capped slider (`IncreaseSlider`) and submit logic (`useSubmitIncrease`), so the arms differ only in presentation.